Inverse variation

1 If y varies inversely with x, write an equation that uses k as the constant of variation.

2 If $(10, 15)$ is on the curve $P = \dfrac{k}{Q}$, solve for k.

3 Determine whether each of the following is an example of a direct variation or an inverse variation:

a The variation relating the distance between two locations on a map and the actual distance between the two locations.

b The variation relating the number of workers hired to build a house and the time required to build the house.

4 If r varies inversely with a, write an equation that uses k as the constant of variation.

5 Determine whether the following equations represent an inverse relationship between x and y:

a $x = 1 + y^3$ **b** $x = \dfrac{8}{y^2}$ **c** $y = 6x + 8$ **d** $xy = -7$

e $x = \dfrac{2}{y}$ **f** $xy = 5x$

6 Consider the equation $s = \dfrac{375}{t}$.

a State the constant of proportionality.

b Find the exact value of s when $t = 6$.

c Find the exact value of s when $t = 12$.

7 State whether the following tables represent an inversely proportional relationship between x and y.

a

x	1	2	3	4
y	3	1.5	1	0.75

b

x	1	2	3	4
y	36	18	12	9

c

x	1	5	6	10
y	3	75	108	300

d

x	1	2	3	4
y	4	5	6	7

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8 m is proportional to $\frac{1}{p}$. Consider the values in the table which represents the relationship.

p	4	6	7	x
m	63	y	36	28

a Determine the constant of proportionality, k.

b Find the values of x and y.

9 Find the equation relating t and s for each of the following tables of values:

a

s	1	2	3	4
t	48	24	16	12

b

s	3	6	9	12
t	$\frac{2}{9}$	$\frac{1}{18}$	$\frac{2}{81}$	$\frac{1}{72}$

10 Use technology to graph the inverse relationship $y = \frac{6}{x}$.

a How many x-intercepts does the graph have?

b How many y-intercepts does the graph have?

Key features of a hyperbola

11 Consider the graph of $y = \frac{2}{x}$.

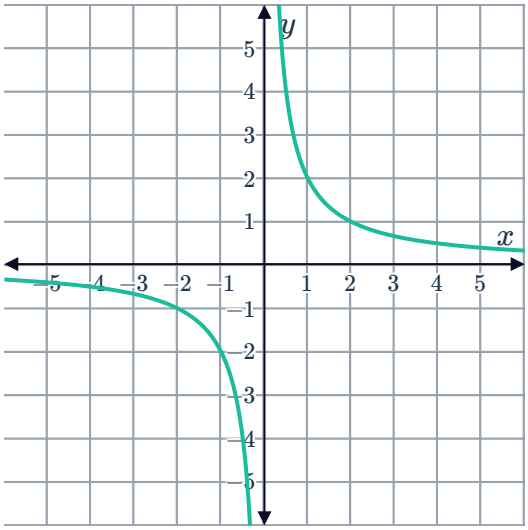
a For $x > 0$, as x increases, what does y approach?

b For $x > 0$, as x approaches 0, what does y approach?

c State the equation of the vertical asymptote.

d State the equation of the horizontal asymptote.

e The graph has two axes of symmetry. State their equations.



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12 The equation $y = -\dfrac{12}{x}$ represents an inverse relationship between x and y.

a When $x = 3$, what is the value of y?

b If x is a positive value, is the corresponding y value positive or negative?

c If x is a negative value, is the corresponding y value positive or negative?

d In which quadrants does the graph of $y = \dfrac{-12}{x}$ lie?

13 Consider the function $y = \dfrac{2}{x}$.

a Complete the table of values.

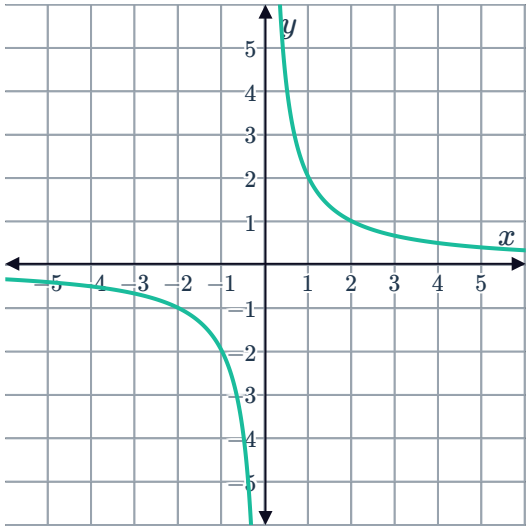
x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{10}$	$-\frac{1}{100}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	1
y									

b For what value of x is the function undefined?

c Rewrite the equation to make x the subject.

d For what value of y is the function undefined?

e Below is the graph of $y = \dfrac{2}{x}$.
Complete the following statement:
As x approaches ▮ from the right, the function value approaches ∞. As x approaches ▮ from the left, the function value approaches $-\infty$.
As x approaches ∞ and $-\infty$, y approaches ▮. This is called the limiting value of the function.



14 Consider the function $y = -\dfrac{5}{x}$.

a For what value of x is the function undefined?

b As x approaches 0 from the positive side, what does y approach?

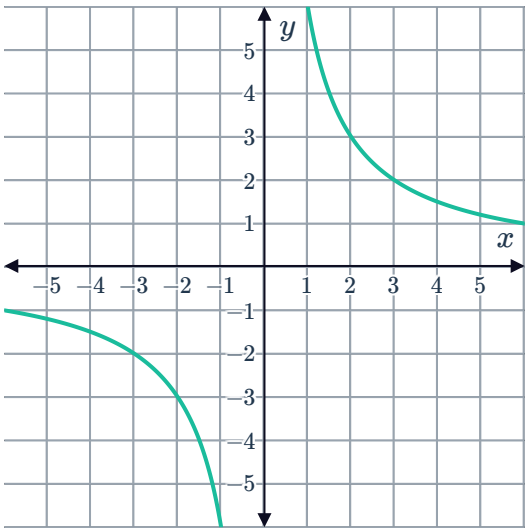
c As x approaches 0 from the negative side, what does y approach?

d As x approaches ∞, what does y approach?

e As x approaches $-\infty$, what does y approach?

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15 Consider the hyperbola shown:
Complete the statement:
"Every point (x, y) on the hyperbola is such that $xy = \square$."

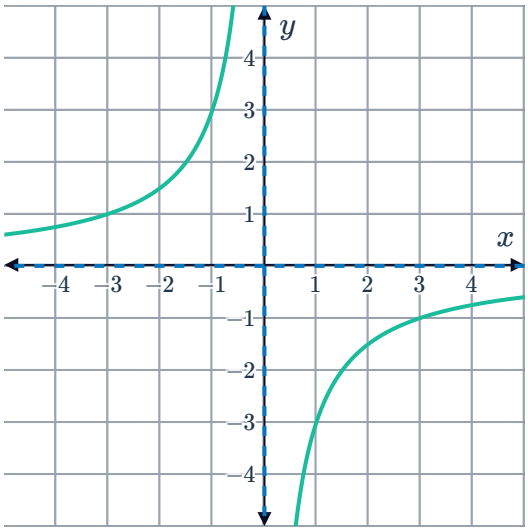


16 Determine whether the following relationships can be modelled by a function of the form $xy = a$?

- **a** The relationship between the number of people working on a job and how long it will take to complete the job.
- **b** The relationship between the number of sales and the amount of revenue.
- **c** The relationship between height and weight.

17 Consider the function graphed below:

- **a** What is the equation of the vertical asymptote?
- **b** What is the equation of the horizontal asymptote?
- **c** State the domain of the function.
- **d** State the range of the function.



18 For functions of the form $f(x) = \dfrac{k}{x}$, where k is a constant, the domain is always $(-\infty, 0) \cup (0, \infty)$. What does this mean for the function when $x = 0$?

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19 Consider the function $y = \frac{1}{x}$.

a For what value of x is the function undefined?

b What is the domain of $y = \frac{1}{x}$?

c For what value of y is the function underfined?

d What is the range of $y = \frac{1}{x}$?

20 Patricia determined the range of the function $y = \frac{2}{x}$ is $(-\infty, 0) \cup (0, \infty)$.
By filling in the gaps below, complete her reasoning.
Looking at $y = \frac{2}{x}$, notice that the numerator ▮ is non-zero, and so $\frac{2}{x}$ can never be equal to ▮.
Another approach is to rearrange the equation and make x the subject. Then we get $x =$ ▮.
In this form we can see that the denominator ▮ cannot be equal to ▮.

21 Consider the functions $y = \frac{4}{x}$ and $y = \frac{2}{x}$.

a When $x = 2$, what is the value of y if $y = \frac{4}{x}$?

b When $x = 2$, what is the value of y if $y = \frac{2}{x}$?

c Which graph lies further away from the axes?

d For hyperbolas of the form $y = \frac{k}{x}$, as the k value increases, what happens to the graph?

Graphs of hyperbolas

22 Consider the inverse variation equation $y = \frac{6}{x}$.

a Complete the following tables of values:

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y					

b Plot the points from the table of values on a cartesian plane.

23 Ursula wants to sketch the graph of $y = \frac{7}{x}$, but knows that it will look similar to many other hyperbolas.
What can she do to the graph to show that it is the hyperbola $y = \frac{7}{x}$, rather than any other hyperbola of the form $y = \frac{k}{x}$?

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24 For each of the following functions:

i Complete a table of values of the form:

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y						

ii Sketch the graph of the function.

iii State which quadrants the function lies in.

a $y = \frac{2}{x}$ **b** $y = -\frac{1}{x}$ **c** $y = -\frac{1}{4x}$

Applications

25 The relationship between the current, C, (in amperes) and resistance, R, (in ohms) in an electrical circuit is given by:

$$C = \frac{240}{R}$$

where the voltage provided to the circuit is 240 V.

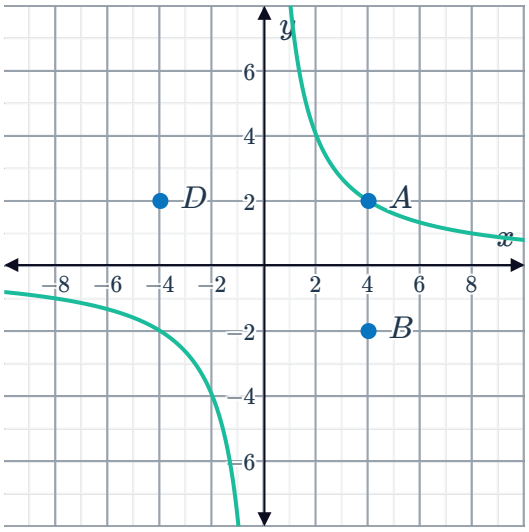
a Graph this function for $0 \leq R \leq 50$.

b What happens to the current as the resistance increases?

26 Consider the hyperbola that has been graphed. Points $A\,(4, 2)$, B, $C\,(-4, k)$ and D form the vertices of a rectangle.

a Find the value of k.

b Hence find the area of the rectangle with vertices $ABCD$.



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27 Boyle's law describes the relationship between pressure and volume of a gas of fixed mass under constant temperature. The pressure for a particular gas can be found using

$$P = \frac{6000}{V}$$

where P has units kg/cm^2 and V has units cm^3.

a Graph the relationship $P = \frac{6000}{V}$ for $0 \leq V \leq 2000$.

b What is the pressure if the volume is $1\ \text{cm}^3$?

c What happens to the pressure as the volume increases?

28 The time it takes a commuter to travel $100\ \text{km}$ depends on how fast they are going. We can write this using the equation $t = \frac{100}{S}$ where S is the speed in km/h and t is the time taken in hours.

a Graph the relationship $t = \frac{100}{S}$.

b What is the time taken if the speed travelled is $10\ \text{km/h}$?

c What is the time taken if the speed travelled is $50\ \text{km/h}$?

d If we want the travel time to decrease, what must happen to the speed of travel?

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